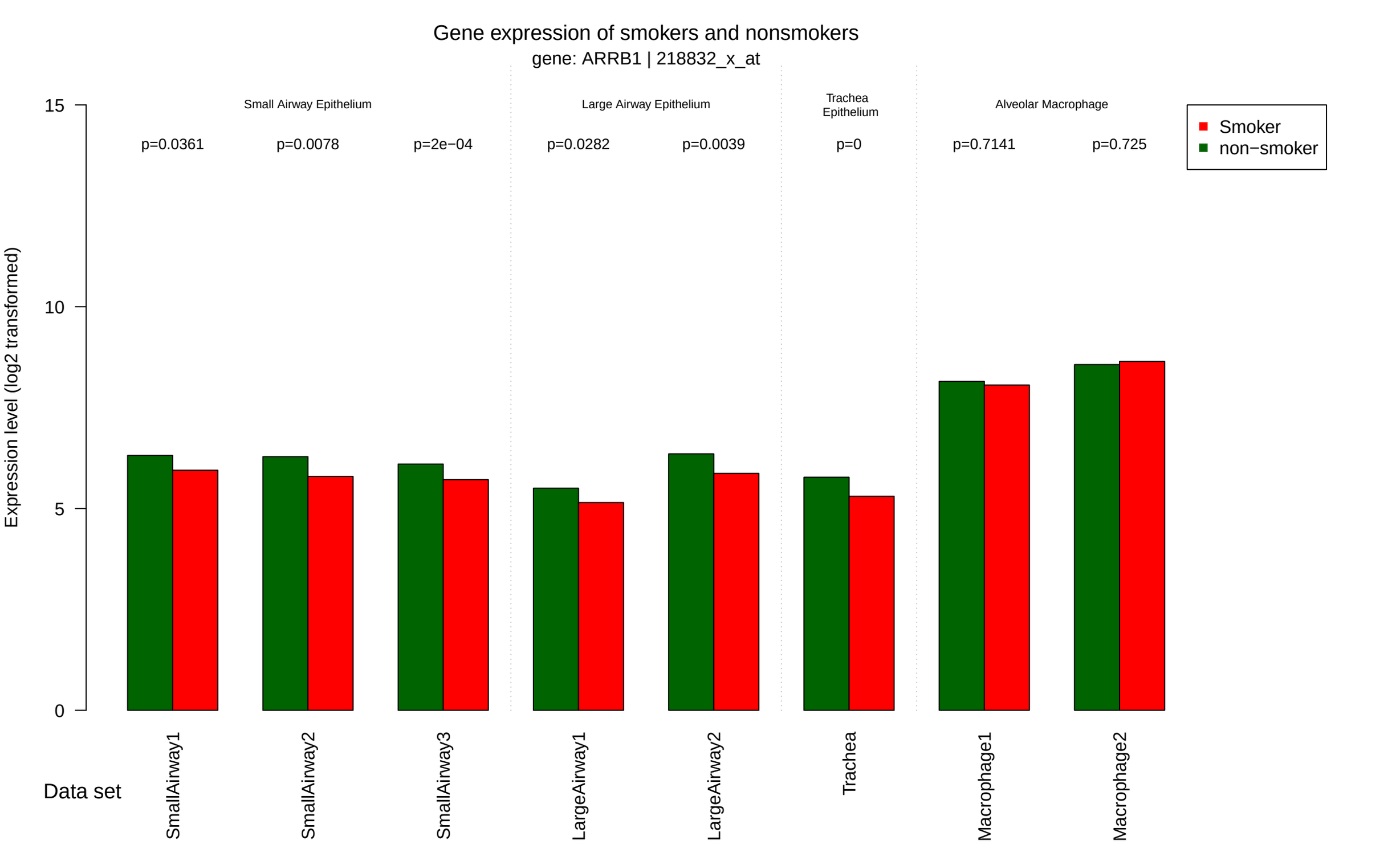
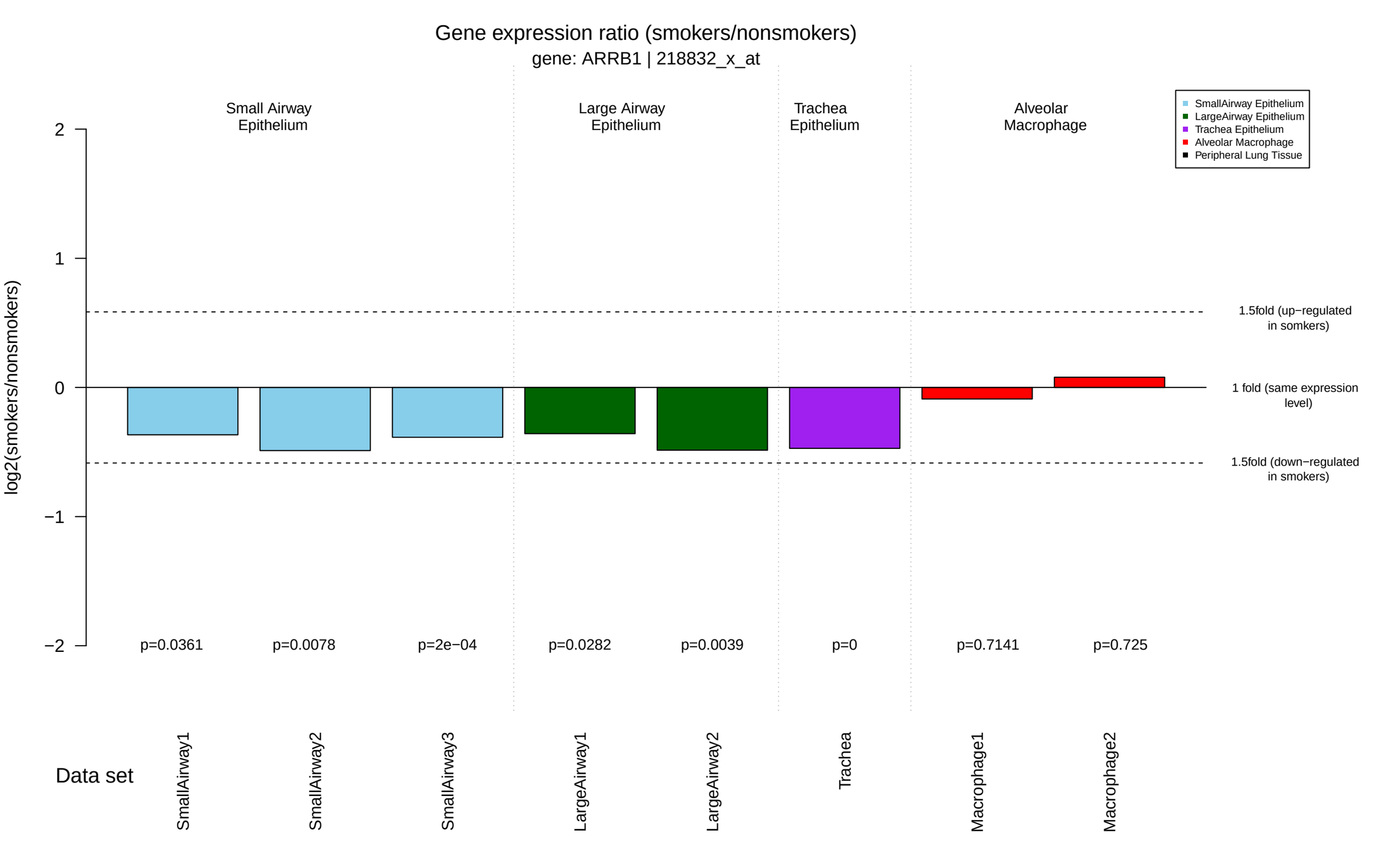
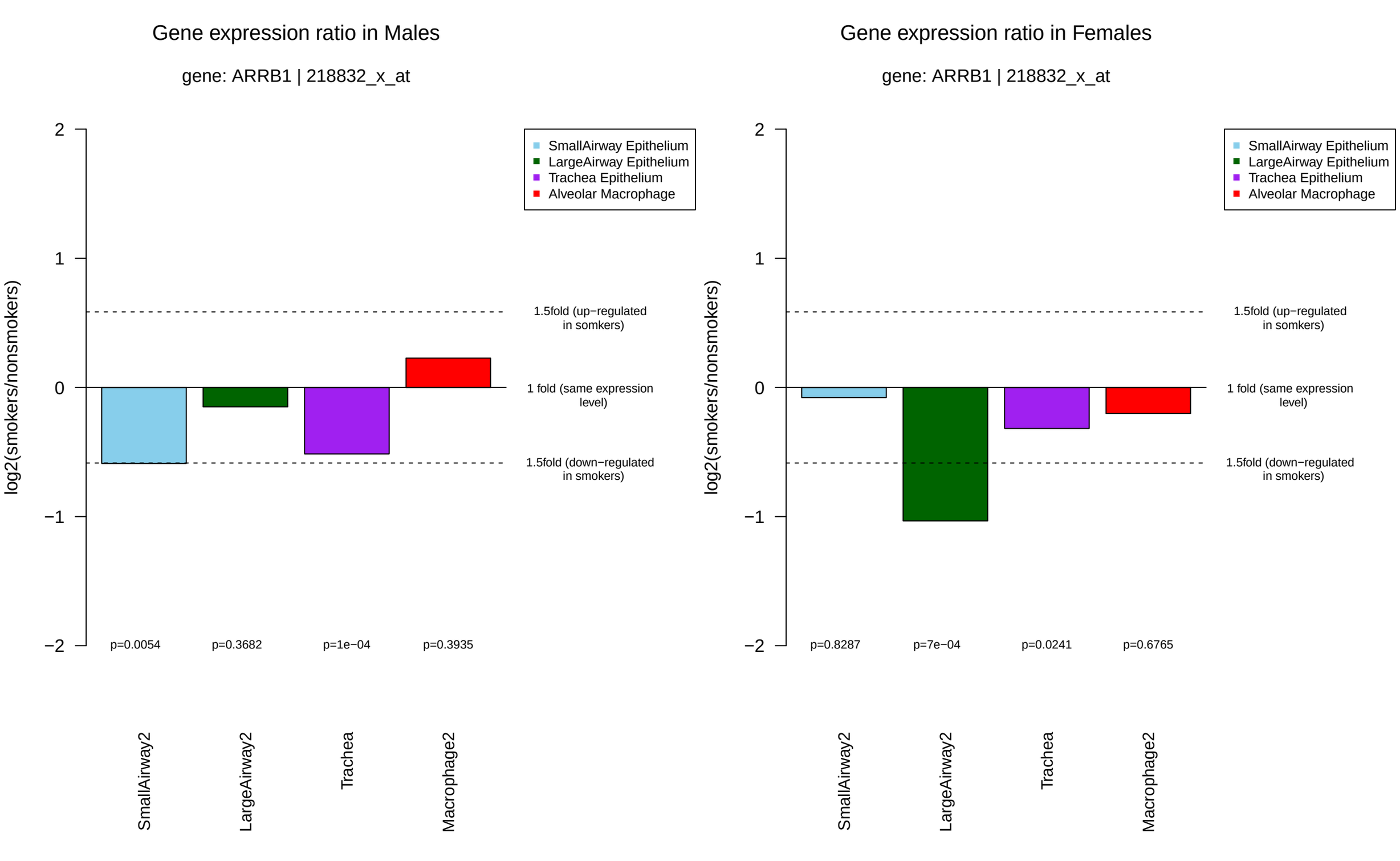
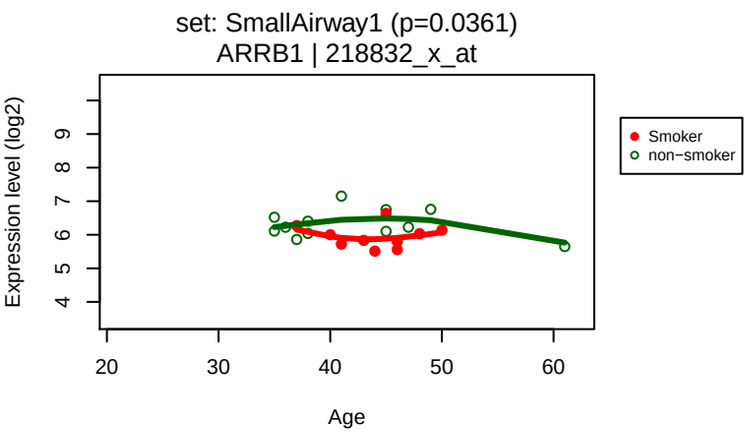
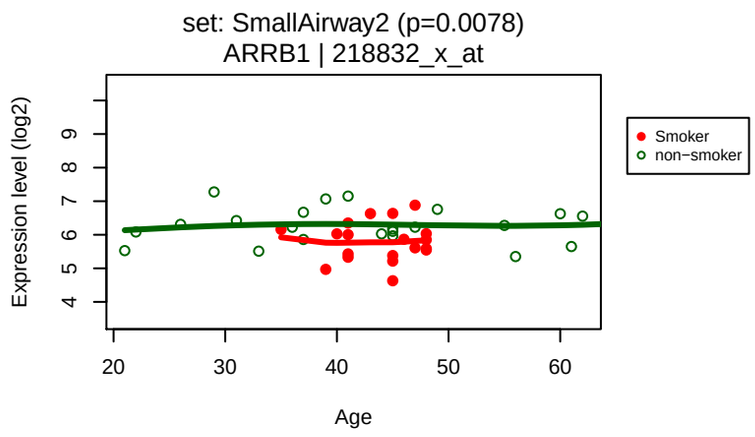
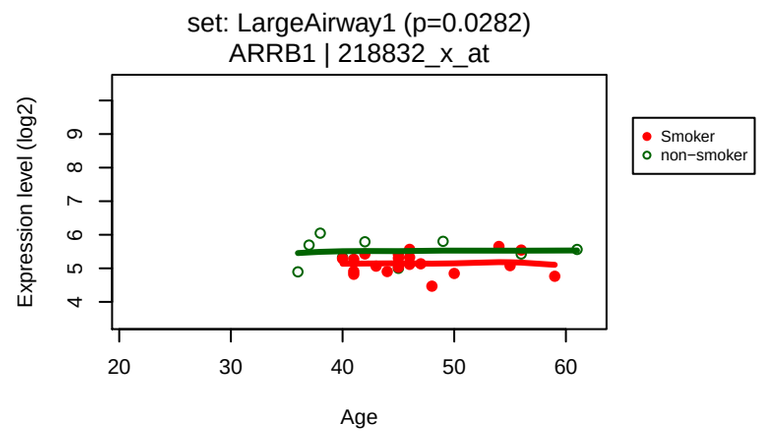

Gene expression of smokers and nonsmokers
gene: ARRB1 | 218832_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0361
p=0.0078
p=2e−04
p=0.0282
p=0.0039
p=0
p=0.7141
p=0.725
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

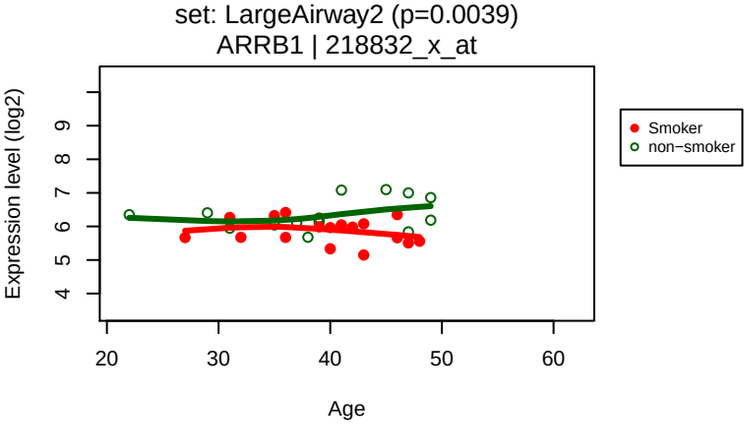

Gene expression ratio (smokers/nonsmokers)
gene: ARRB1 | 218832_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0361
p=0.0078
p=2e−04
p=0.0282
p=0.0039
p=0
p=0.7141
p=0.725
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

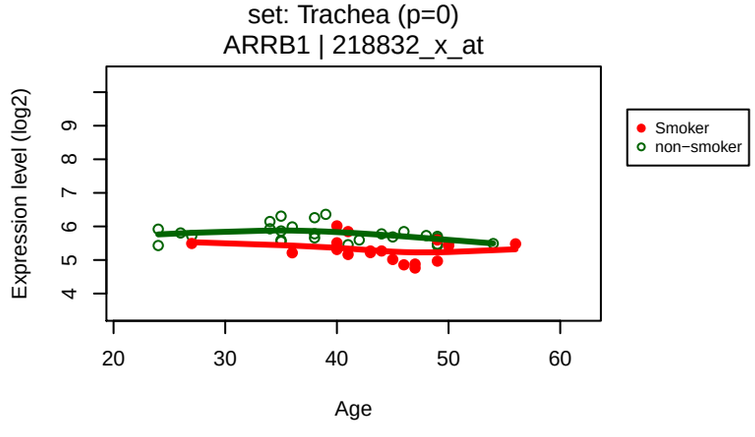

Gene expression ratio in Males
gene: ARRB1 | 218832_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0054
p=0.3682
p=1e−04
p=0.3935
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARRB1 | 218832_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8287
p=7e−04
p=0.0241
p=0.6765
SmallAirway2
LargeAirway2
Trachea
Macrophage2

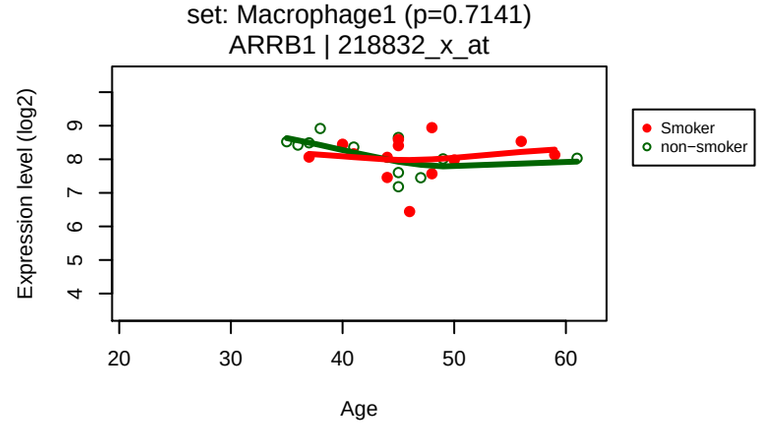

set: SmallAirway1 (p=0.0361)
ARRB1 | 218832_x_at
Expression level (log2)
Age
Smoker
non−smoker

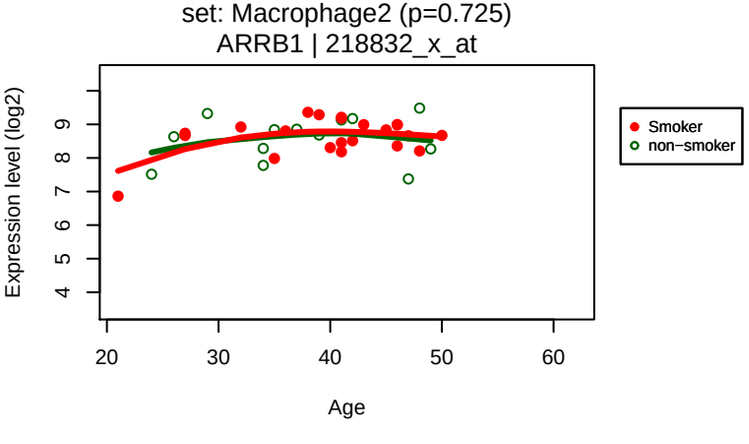

set: SmallAirway2 (p=0.0078)
ARRB1 | 218832_x_at
Expression level (log2)
Age
Smoker
non−smoker

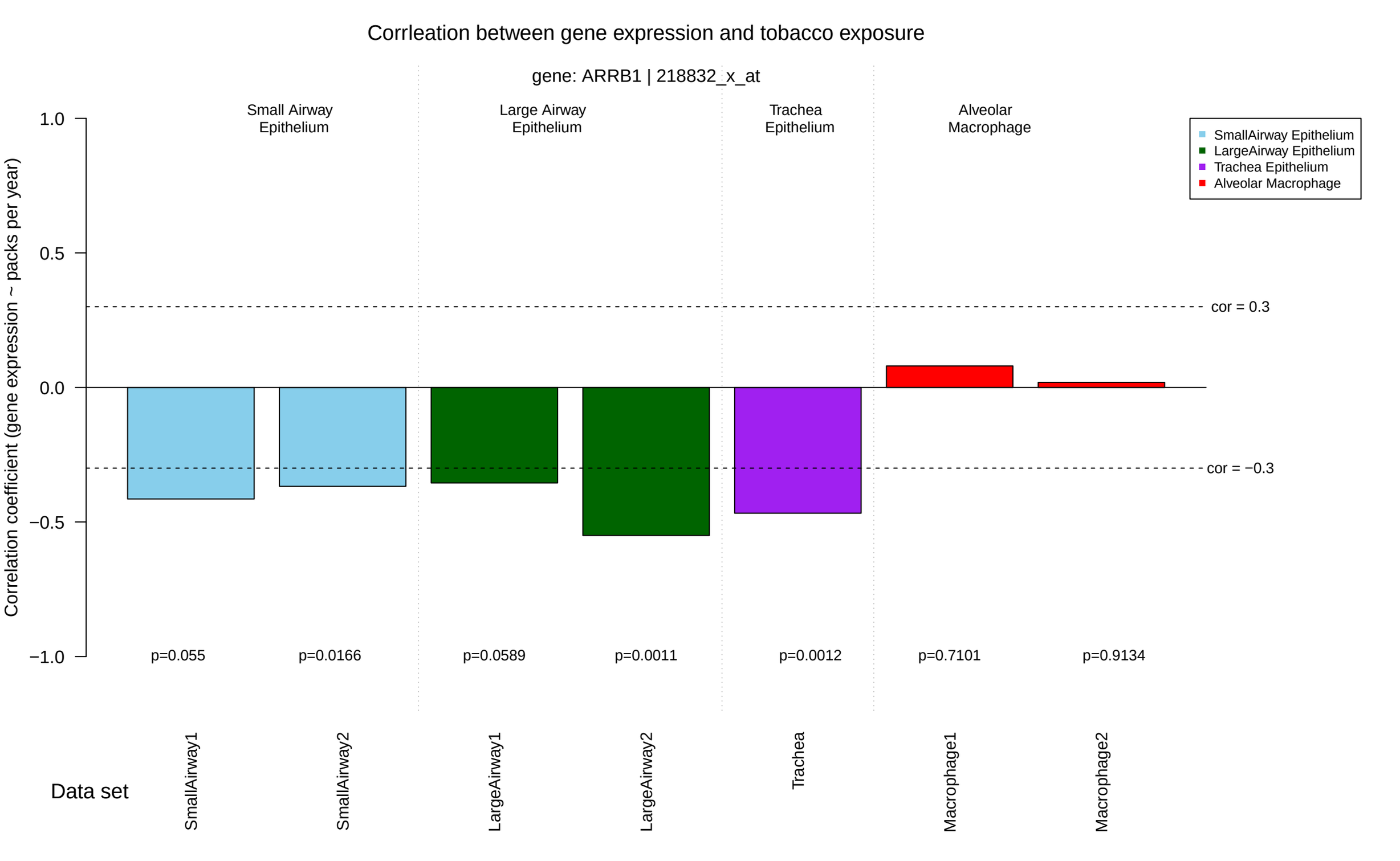

set: LargeAirway1 (p=0.0282)
ARRB1 | 218832_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0039)
ARRB1 | 218832_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
ARRB1 | 218832_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7141)
ARRB1 | 218832_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.725)
ARRB1 | 218832_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARRB1 | 218832_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.055
p=0.0166
p=0.0589
p=0.0011
p=0.0012
p=0.7101
p=0.9134
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2